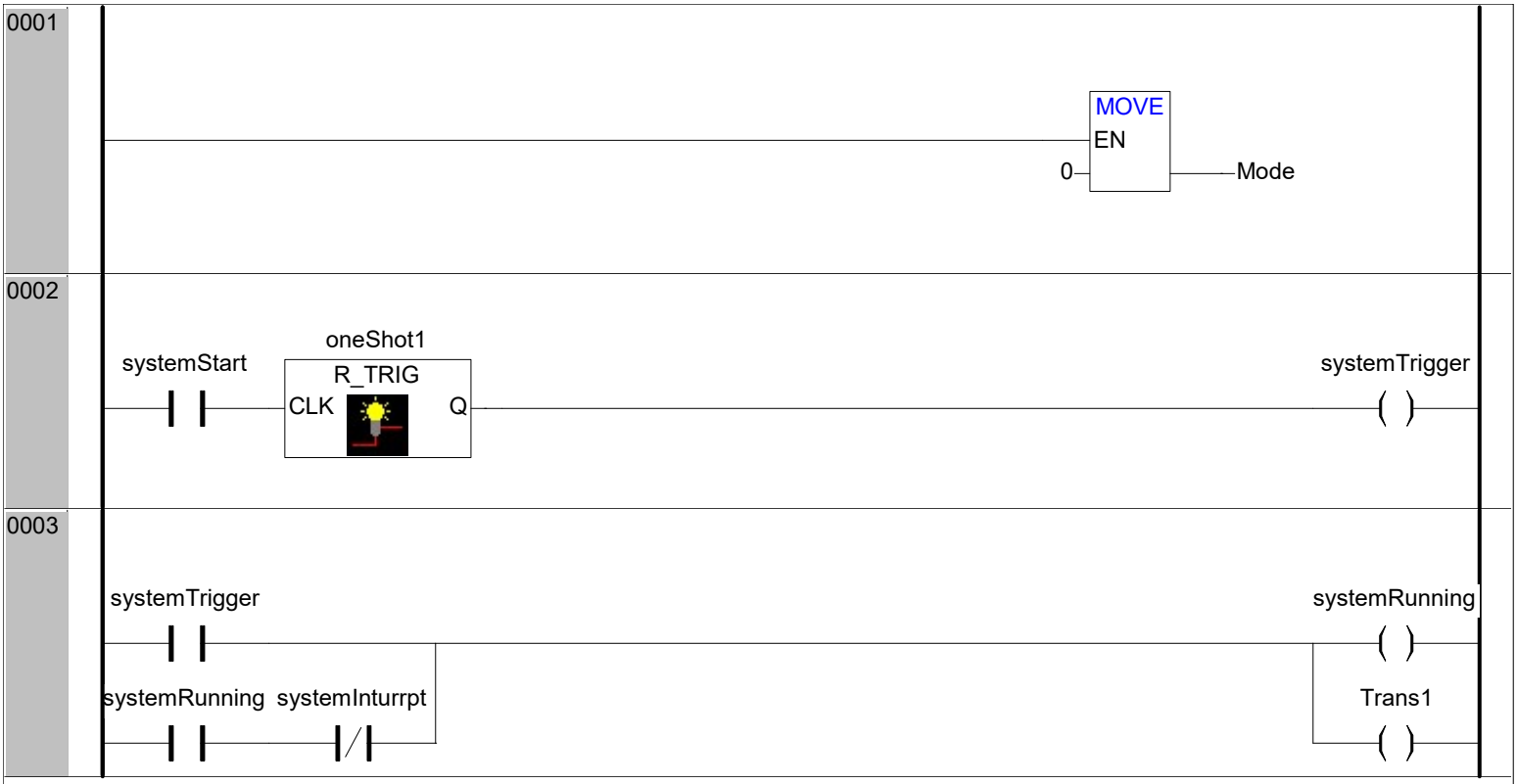
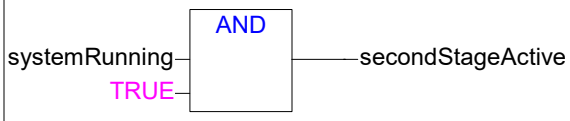
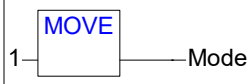
```
VAR_GLOBAL
    systemRunning:BOOL;
    LVlInput:DINT;
    TempInput:DINT;
    LvlRes:REAL;
    TempRes:REAL;
    Pump:BOOL;
    Mixer:BOOL;
    Valve:BOOL;
    Heater:BOOL;
    DrainRate:DINT;
    FillRate:DINT;
    HeatUp:DINT;
    DropHeat:DINT;
    HHTempAlm:BOOL;
    LLLvlAlm:BOOL;
    HHLvlAlm:BOOL;
    secOP:TIME;
    minOP:DINT;
    hourOP:DINT;
    Mode:DINT;
END_VAR
```


0001
MOVE
EN
0
Mode
0002
systemStart
oneShot1
R_TRIG
CLK
Q
systemTrigger
0003
systemTrigger
systemRunning
systemRunning
systemInturrpt
Trans1

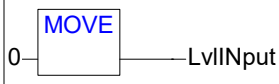
0001



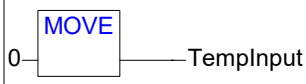
0002



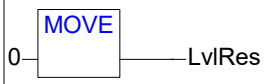
0003



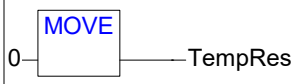
0004



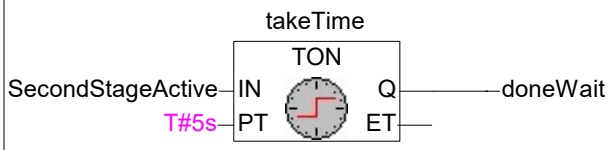
0005



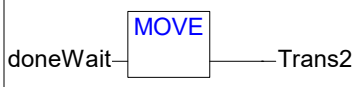
0006



0007



0008



```
FIllRate:=2;
DrainRate:=1;
HeatUp:=2;
DropHeat:=1;
Mode:=2;

IF systemRunning  THEN
    WHILE LvlRes < 69 DO
        Time3.IN:=TRUE;
        Time3.PT:=T#500ms;
        Time3;

        LvlRes:= LvlRes+FillRate;
        TempRes:= TempRes-DropHeat;

        IF LvlRes < 40  THEN
            Time1.IN:=TRUE;
            Time1.PT:=T#5s;
            Time1;

            ons2.CLK:=TIme1.Q;
            ons2;

            Pump:=ons2.Q;

        ELSE
            Time2.IN:=TRUE;
            Time2.PT:=T#5s;
            Time2;

            Pump:=0;
            EXIT;
        END_IF
        TIme3.IN:=FALSE;
        Time3;

    END_WHILE

WHILE TempRes<350 DO
        Time4.IN:=TRUE;
        Time4.PT:=T#500ms;
        Time4;

        TempRes:= TempRes+HeatUp;

        IF TempRes < 300  THEN
            Time5.IN:=TRUE;
            Time5.PT:=T#5s;
            Time5;

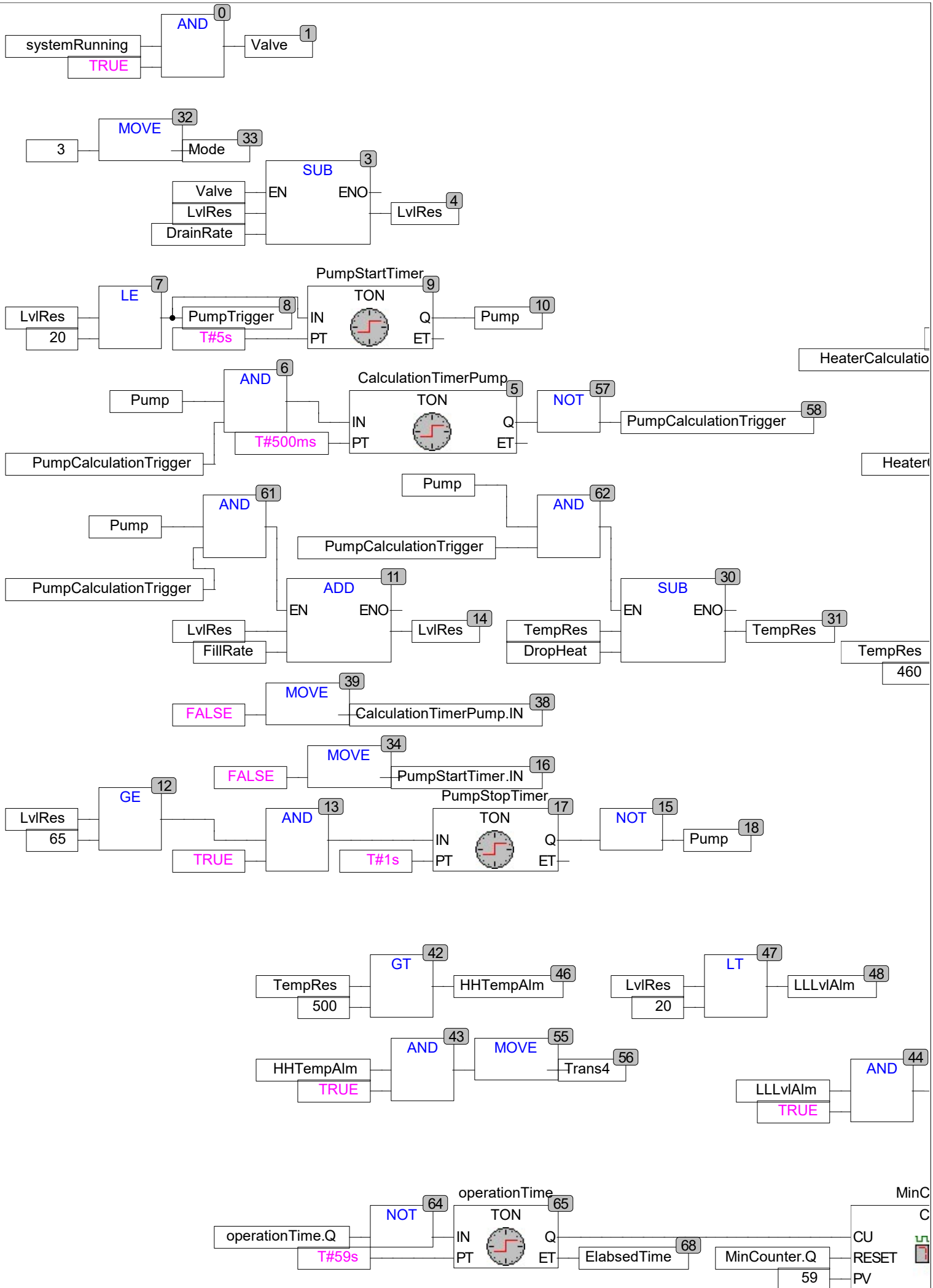
            ons3.CLK:=TIme1.Q;
            ons3;

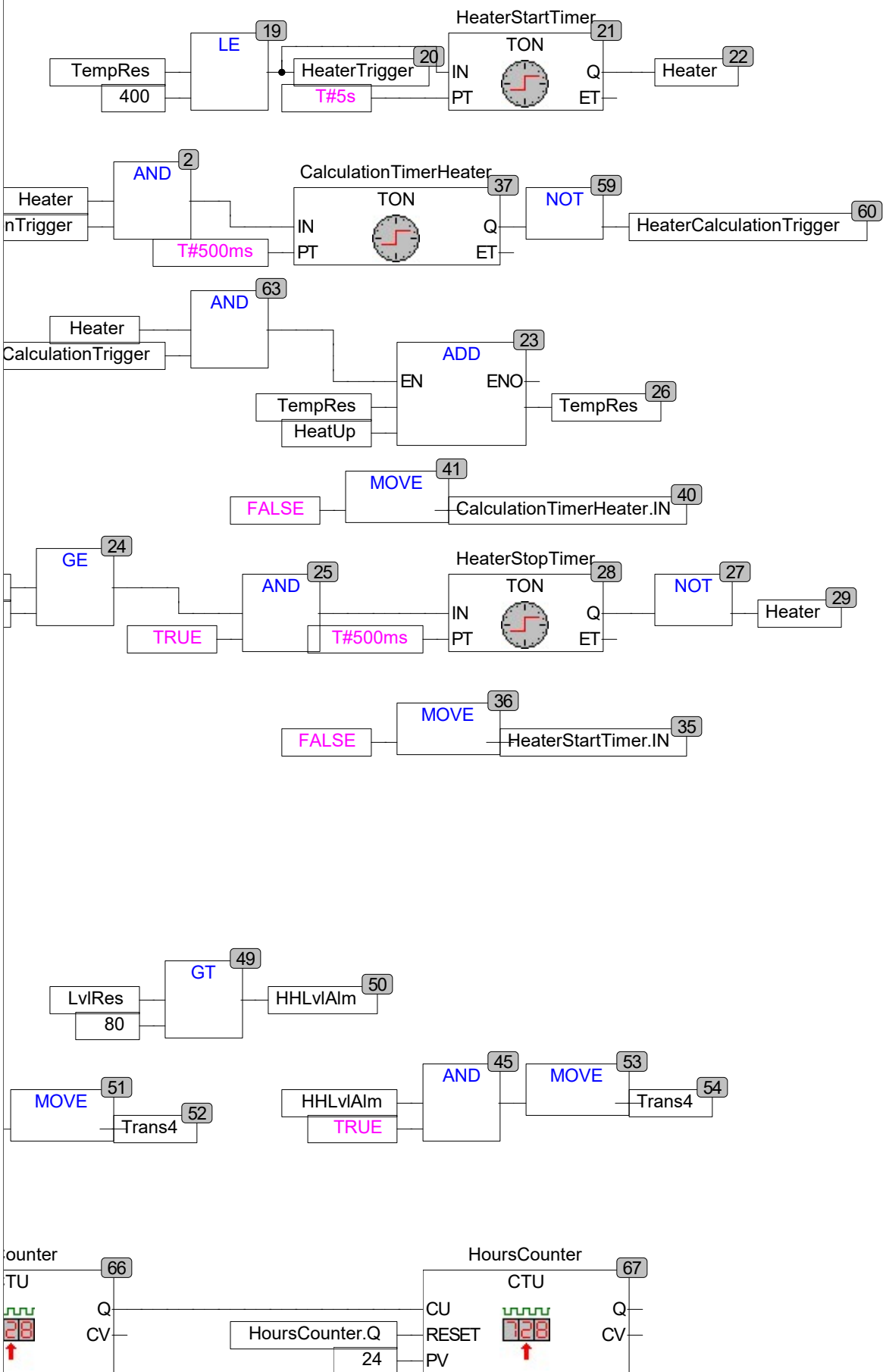
            Heater:=ons2.Q;
        ELSE
            Time6.IN:=TRUE;
            Time6.PT:=T#5s;
            Time6;

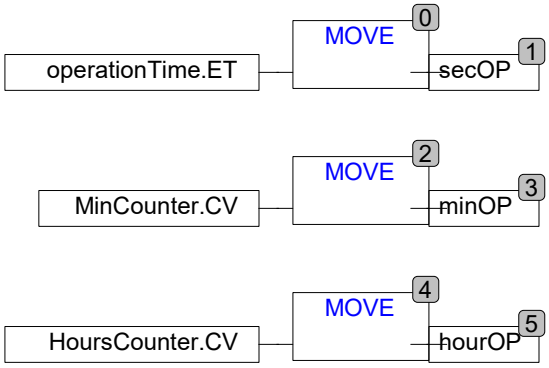
            Heater:=0;
            EXIT;
        END_IF
        TIme4.IN:=FALSE;
        Time4;
    END_WHILE
```

```
0069        WHILE Heater=0 AND pump=0 DO
0070            Time7.IN:=TRUE;
0071            Time7.PT:=T#500ms;
0072            Time7;
0073
0074            TempRes:= TempRes-DropHeat;
0075
0076            IF TempRes > 300  THEN
0077                Time8.IN:=TRUE;
0078                Time8.PT:=T#5s;
0079                Time8;
0080
0081                ons4.CLK:=TIme1.Q;
0082                ons4;
0083
0084                Mixer:=ons2.Q;
0085            ELSIF TempRes < 300 THEN
0086                Time9.IN:=TRUE;
0087                Time9.PT:=T#30s;
0088                Time9;
0089
0090                Mixer:=0;
0091
0092                EXIT;
0093
0094            END_IF
0095            TIme7.IN:=FALSE;
0096            Time7;
0097        END_WHILE
0098
0099
0100        IF LvlRes > 40 AND LvlRes<70  AND TempRes>298 THEN
0101                Trans3:= 1;
0102        END_IF
0103    END_IF
```

```
0001 Counter1.PV:=999;
0002 Counter1.CU:=secondStageActive;
0003 secondStageCount:= Counter1.CV;
0004
0005 Counter1;
```


systemRunning
TRUE
AND
0
Valve
1
3
MOVE
32
Mode
33
Valve
LvlRes
DrainRate
SUB
3
EN
ENO
LvlRes
4
LvlRes
20
LE
7
PumpTrigger
8
T#5s
PumpStartTimer
TON
9
IN
PT
Q
ET
Pump
10
HeaterCalculatio
Pump
PumpCalculationTrigger
AND
6
T#500ms
CalculationTimerPump
TON
5
NOT
57
PumpCalculationTrigger
58
Heater(
Pump
AND
61
PumpCalculationTrigger
Pump
AND
62
PumpCalculationTrigger
ADD
11
LvlRes
FillRate
LvlRes
14
SUB
30
TempRes
DropHeat
TempRes
31
TempRes
460
FALSE
MOVE
39
CalculationTimerPump.IN
38
FALSE
MOVE
34
PumpStartTimer.IN
16
LvlRes
65
GE
12
TRUE
AND
13
T#1s
PumpStopTimer
TON
17
NOT
15
Pump
18
TempRes
500
GT
42
HHTempAlm
46
LvlRes
20
LT
47
LLLvlAlm
48
HHTempAlm
TRUE
AND
43
MOVE
55
Trans4
56
LLLvlAlm
TRUE
AND
44
operationTime.Q
T#59s
NOT
64
operationTime
TON
65
ElabsedTime
68
MinCounter.Q
59
MinC
C
CU
RESET
PV


HeaterStartTimer
19
LE
TempRes
400
20
HeaterTrigger
T#5s
TON
21
IN
PT
Q
ET
Heater
22
2
AND
Heater
nTrigger
CalculationTimerHeater
37
TON
IN
PT
Q
ET
NOT
59
HeaterCalculationTrigger
60
T#500ms
63
AND
Heater
CalculationTrigger
23
ADD
EN
ENO
TempRes
HeatUp
TempRes
26
41
MOVE
FALSE
CalculationTimerHeater.IN
40
24
GE
25
AND
TRUE
T#500ms
HeaterStopTimer
28
TON
IN
PT
Q
ET
NOT
27
Heater
29
36
MOVE
FALSE
HeaterStartTimer.IN
35
49
GT
LvlRes
80
HHLvlAlm
50
51
MOVE
Trans4
52
HHLvlAlm
TRUE
45
AND
53
MOVE
Trans4
54
ounter
TU
66
Q
CV
HoursCounter.Q
24
HoursCounter
67
CTU
CU
RESET
PV
Q
CV


MOVE
0
operationTime.ET
secOP
1
MOVE
2
MinCounter.CV
minOP
3
MOVE
4
HoursCounter.CV
hourOP
5

```
0001      LD      0
0002      ST      Valve
0003
0004      LD      4
0005      ST      Mode
0006
0007      LD      LvlRes
0008      GT      80
0009      JMPC    HHLvlAlarm
0010
0011
0012      LD      LvlRes
0013      LT      20
0014      JMPC    LLLvlAlarm
0015
0016
0017
0018      LD      TempRes
0019      GT      500
0020      JMPC    HHTempAlarm
0021
0022
0023      LD      T#10s
0024      ST      Time10.PT
0025      LD      TRUE
0026      ST      TIme10.IN
0027      CAL         Time10
0028      LD      0
0029      ST      Heater
0030      LD      0
0031      ST      Pump
0032      LD      0
0033      ST      Mixer
0034      LD      0
0035      ST      Valve
0036
0037      LD      FALSE
0038      ST      Time10.IN
0039      CAL         Time10
0040
0041      LD      whereToGo
0042      EQ      1
0043      JMPC    EndProgram1
0044
0045      LD      whereToGo
0046      EQ      0
0047      JMPC    EndProgram2
0048
0049  HHLvlAlarm:
0050      LD      1
0051      ST      Valve
0052      LD      0
0053      ST      Pump
0054      LD      LvlRes
0055      SUB         DrainRate
0056      ST      LvlRes
0057      RET
0058
0059  LLLvlAlarm:
0060      LD      0
0061      ST      Valve
0062      LD      1
0063      ST      Pump
0064      LD      LvlRes
0065      ADD         FillRate
```

```
0069 HHTempAlarm:
0070     LD      0
0071     ST      Heater
0072     LD      1
0073     ST      Pump
0074     LD      1
0075     ST      Mixer
0076     LD      1
0077     ST      Valve
0078     LD      TempRes
0079     SUB         DropHeat
0080     SUB         DropHeat
0081     ST      TempRes
0082     LD      LvlRes
0083     ADD         FillRate
0084     SUB         DrainRate
0085     ST      LvlRes
0086     RET
0087
0088 EndProgram1:
0089
0090     LD      1
0091     ST      Trans0
0092
0093 ENdProgram2:
0094
0095     LD      1
0096     ST      Trans5
0097
```